

two or more hidden layers are known as deep neural networks. A historical note on the evolution of neural networks can classify them as belonging to one of the three generations. This classification is based on the computational ability of the individual neuron used in the neural network.

The first generation of neural networks was based on neurons that were threshold gates or perceptrons. These neurons did not have a non-linear activation function; instead, their output was either 1 or 0, depending on whether the weighted sum of their inputs was above or below a threshold 't'. An important feature of these neural networks is that they can only produce digital outputs. However, they are capable of modelling every Boolean function and hence are universal for computation with digital inputs and outputs.

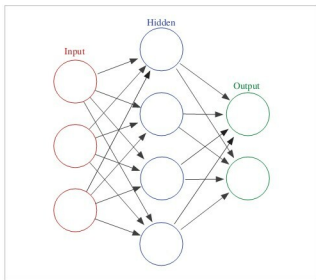


Figure 2: A simple neural network

The second generation of neural networks consists of neurons which apply a non-linear continuous function to the sum of weighted inputs, and hence produce a continuous set of possible output values. Some of the examples of activation functions are the sigmoidal and tanh functions. A typical example of second generation neural networks is the feed forward neural network such as the one shown in Figure 2, which we discussed earlier. Other examples of second generation neural networks include recurrent neural networks and recursive neural networks, which we will discuss in subsequent columns.

Feed forward neural networks are those in which signals travel only in one direction, i.e., from input to output. In other words, the output of an earlier layer in the architecture feeds only to later layers and not the other way around. The output of any layer is not fed to itself or any other earlier layers in feed forward neural networks. The example shown in Figure 2 is a feed forward neural network. On the other hand, recurrent neural networks can have the output of one layer being fed back through the network into an earlier layer, over time.

Essentially, this means that the neural network has loops or cycles in which it allows the output computed at layer 'i' at time 't' being fed back to an earlier layer at time 't+1'. This allows the recurrent neural networks to achieve some form of short-term memory. While feed forward neural networks are used for single input value classification, recurrent neural networks are typically used for sequence classification and time series classification.

Second generation neural networks can also support digital outputs by applying a threshold at the output, and hence can model any arbitrary Boolean function. They are also universal for analogue computations, since any continuous function can be approximated reasonably well by means of a second generation neural network with a single hidden layer itself.

Today, these second generation neural networks are widely used in various applications such as speech processing, image recognition and natural language processing. These neural networks are typically trained using gradient descent algorithms and back propagation. In order to apply neural networks to different tasks, one of the important challenges is how to effectively and quickly train these neural networks. In fact, one of the reasons for the resurgent popularity of artificial neural networks is that they can be trained efficiently using GPUs. We will discuss more about training neural networks in subsequent columns.

The third generation of neural networks employs spiking neurons, also known as 'integrate and fire' neurons. These neural networks more closely model the biological neurons' activity compared to first and second generation neurons. These spiking neural networks can also be used for information processing, just as the second generation of neural networks. While their effectiveness has been described theoretically, they have not yet been widely used in real life applications due to their high computational demands. Hence, we will not discuss spiking neural networks, except to note that there is considerable research happening in this field. For instance, Qualcomm recently announced that Qualcomm Zeroth processors are based on the principle of spiking neural networks, which is an example of neuromorphic architectures—otherwise known as biologically inspired computing architectures.

We will focus mainly on the second generation neural networks in this and subsequent columns, since large real life natural language processing applications have been built using them.

The main application of neural networks in information processing is to use them as classifiers. In traditional machine learning, different classifiers such as logistic regression classifiers and support vector machines are popularly used to classify text, images or speech. Similarly, fully connected neural networks can be used in place of traditional ML classifiers such as SVM, as a drop-in replacement for them. In many cases, neural network based classification has been